Statistics Final Project

Anastasia Dobrynina

Paper:

Esipova, M. (**2019**). **Composition and projection of co-speech gestures**. In *Semantics and Linguistic Theory* (pp. 117-137). DOI: https://doi.org/10.3765/salt.v29i0.4600

Introduction

Esipova (2019) investigates the semantic composition and projection of content-bearing co-speech gestures. While previous literature assumed that gestures project their meaning in a single, uniform way based solely on their co-speech status, Maria Esipova hypothesizes that a gesture's interpretation actually depends on its structural role within a sentence. Specifically, the research tests whether gestures can compose as either **restricting modifiers**, **non-restricting modifiers**, or **supplements**, just like spoken words. To prove this, the study seeks to compare the acceptability of gestures against regular spoken **adjectives** and **appositives** across different semantic contexts.

Data description

To test these predictions, an online acceptability judgment experiment was conducted on Amazon Mechanical Turk. The final sample consists of 122 native English speakers (33 female, 89 male), established after excluding respondents who failed an embedded attention check or reported speaking other native languages. The study utilized a within-subjects design manipulating two primary variables: **Content Type** and **Interpretation** context. Participants watched videos of sentences with these target expressions and rated their naturalness using a 0–100 pseudo-continuous slider. Finally, the resulting dataset is complete; a direct check of the raw data file in R confirms there are exactly 0 missing values, as participants were required to evaluate all 11 assigned items (9 test items and 2 additional checks) to complete the survey

Exploratory Data Analysis

Data Description & Sample Characteristics

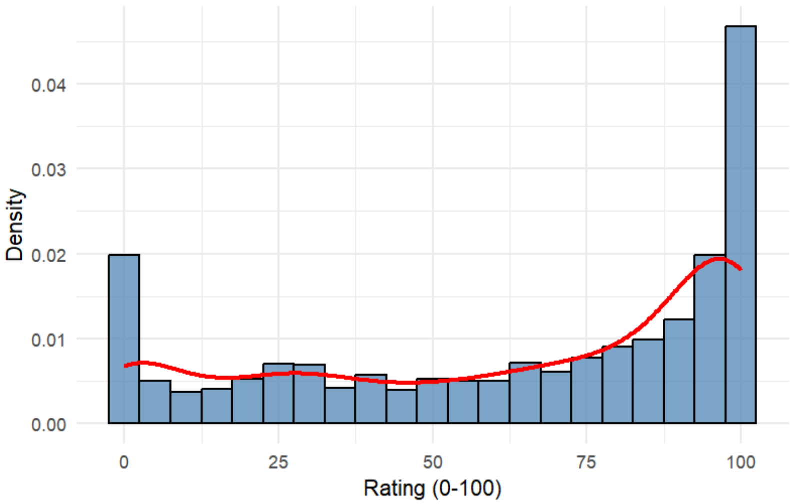
The analyzed dataset contains a total of 1,220 observations across 11 variables. The primary dependent variable is the Rating (a numeric variable ranging from 0 to 100). The main independent variables are Content (categorical: Adjective, Appositive, Gesture) and Context (categorical: PNR, R, NPNR). Other structural variables include Participant and Scenario. A formal check confirmed that there are no missing values (0 NAs) in the final dataset, indicating complete responses from all 122 participants.

Visualizations and Data Quality

1. Histogram and Density Plot

The distribution of overall ratings reveals a strongly **non-normal, bimodal pattern**, with responses heavily clustered near the extremes (100 and 0). This ceiling and floor effect is typical for bounded slider-scale acceptability judgments.

Figure 1. Distribution of Acceptability Ratings



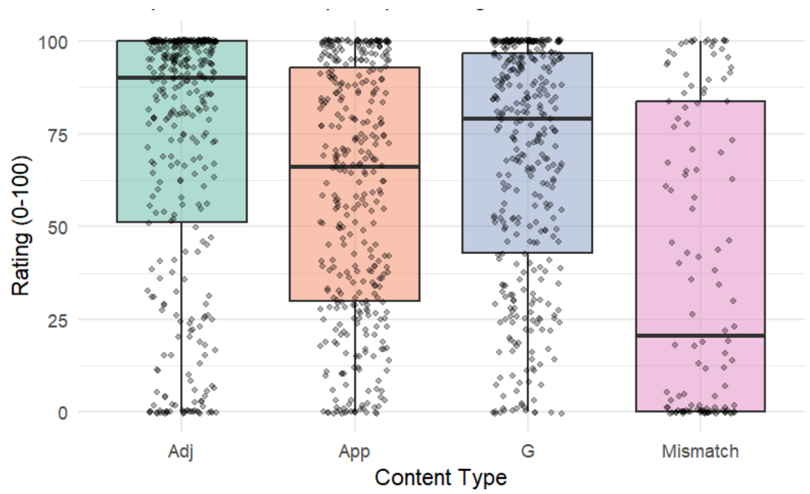
2. Boxplots by Content

This plot shows the ratings for Adjectives, Appositives, Gestures, and Mismatch. Here, we group the data only by Content Type and ignore the different contexts. Because we mix good contexts and bad contexts together in one group, the data spreads out very widely —- the whiskers touch the top (100) and bottom (0) of the scale. This suggests that **we cannot analyze Content Type alone** —- we must also include Context to understand the real patterns. The dots show strong polarization: most answers cluster at the extremes.

The **Mismatch condition** successfully acts as a baseline for unacceptable utterances, showing the lowest overall mean (M = 39.0). However, there is a cluster of data points near 100 in this condition. Most likely, these participants ignored the written context. Nevertheless, we do not exclude these observations from the dataset because the pre-defined exclusion criteria relied solely on a specific attention check question, and removing "inconvenient" data points post-hoc would constitute severe data manipulation.

An interesting anomaly is observed in the **behavior of appositives and gestures**. Theoretically, the NPNR interpretation represents an unnatural context, so we expect very low acceptability ratings. Adjectives behave exactly as predicted, dropping almost to the level of the Mismatch baseline (M = 42.4, Mdn = 29). However, both appositives (M = 52.4, Mdn = 55) and gestures (M = 50.5, Mdn = 51) are surprisingly tolerated, gathering ratings around the middle of the scale. The exact theoretical reasons why these two content types are not strongly dispreferred remains unclear.

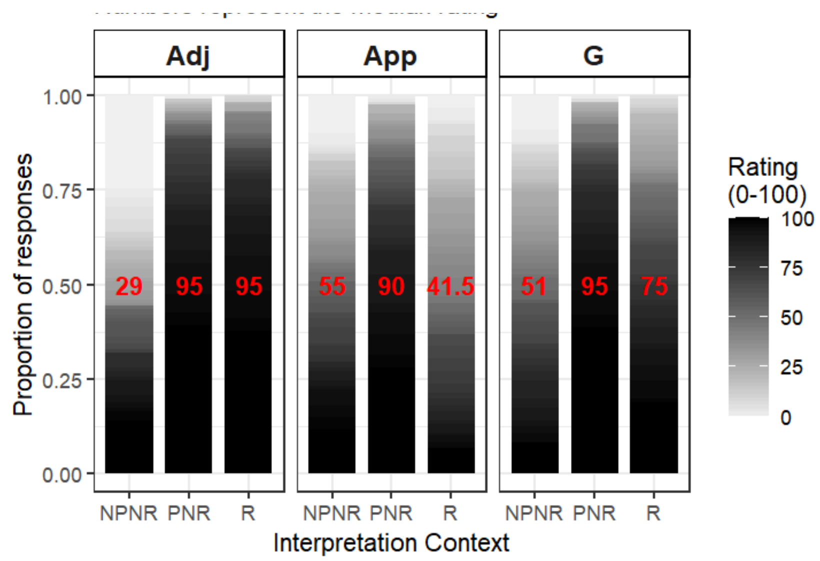
Figure 2. Acceptability Ratings by Content Type (Dots represent individual participant ratings)



3. Proportional Stacked Bar Charts

This chart shows exactly how all participants answered. The red numbers show the median score. The vertical axis shows the percentage of people. The color rule is simple: the darker the color, the higher the rating. However, this plot is not just about colors. It is about the distribution of answers. It helps us see the full picture: did people agree and give similar scores, or did their opinions split into completely different groups?

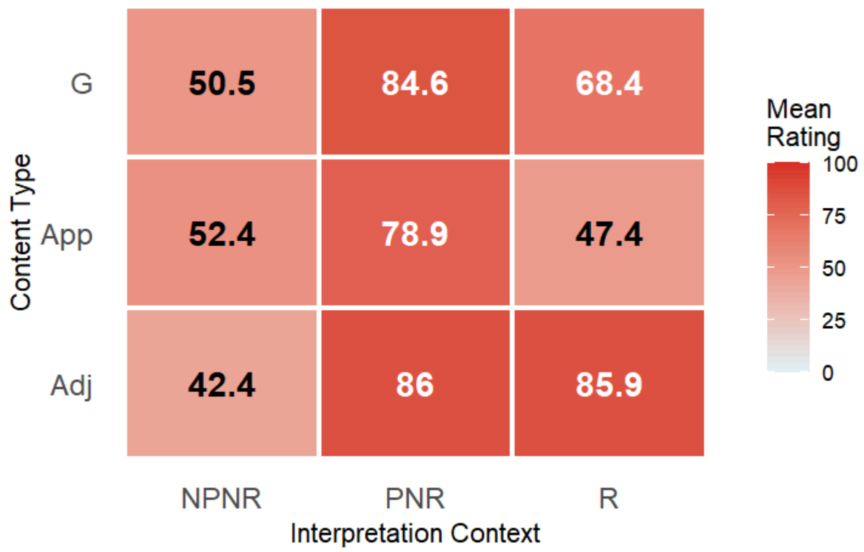
Figure 3. Interaction of Context and Content on Ratings (Numbers represent the median rating)



4. Heatmap of Mean Ratings

The heatmap visualizes the mean acceptability ratings for each combination of Content Type and Interpretation Context. The exact mean values are also printed inside each tile. This format provides a clean and immediate view of the **overall interaction effect between the two variables** without splitting the data into separate panels. This is crucial for our interaction, therefore this is the best visualization among others.

Figure 4. Heatmap: Interaction of Content and Context (Colors and numbers represent mean acceptability ratings)



Descriptive Statistics

Table 1. Basic Descriptive Statistics

Content Type	N	Mean Rating	SD	Median
Adjectives	366	71.4	35.4	90.0
Appositives	366	59.5	33.6	66.0
Gestures	366	67.8	31.8	79.0
Mismatch	122	39.0	40.2	20.5

Adjectives received the highest overall mean acceptability rating (M = 71.4, SD = 35.4), followed closely by Gestures (M = 67.8, SD = 31.8), and Appositives (M = 59.5, SD = 33.6). The Mismatch baseline yielded the lowest overall mean (M = 39.0, SD = 40.2). Looking deeper into the critical Gesture group, the Projecting Non-Restricting (PNR) context yielded the highest mean rating (M = 84.6, SD = 22.6), Restricting (R) yielded a moderate mean (M = 68.4, SD = 28.8), and Non-Projecting Non-Restricting (NPNR) was the lowest (M = 50.5, SD = 33.5). This mathematical breakdown perfectly aligns with the visual analysis and Esipova's theoretical predictions.

Outliers, Patterns, and Data Quality

The analysis confirms high data quality alongside expected experimental noise. Using the IQR method, exactly 39 mathematical outliers were identified. A check for data quality revealed 33 instances where participants gave suspiciously high ratings (over 80) to the absurd Mismatch baseline, likely due to a lack of attention to the written context. However, a variance check is very encouraging: there were 0 participants with zero variance, meaning every respondent actively engaged with the survey rather than simply clicking the same rating for every trial.

Descriptive Statistics Summary

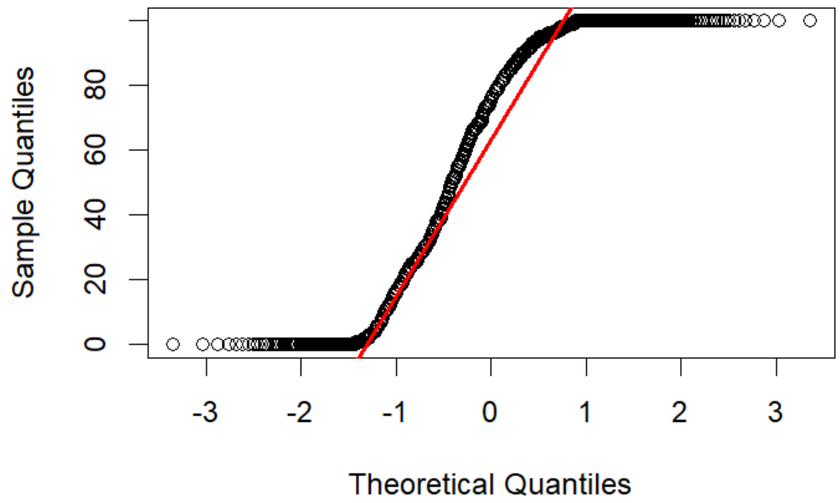
These statistics above highlight the necessity of considering the experimental Context to explain the observed variance. Given that the mean ratings are heavily influenced by polarized responses and high standard deviations, simple descriptive statistics are insufficient to confirm the study's hypotheses. This necessitates the use of inferential statistical models to account for these patterns more formally.

Statistical Assumptions Check

Before proceeding to inferential models, we tested the core assumptions of normality and homogeneity of variance.

The visual inspection of the Q-Q plot shows severe deviations from the diagonal reference line at both ends of the scale, confirming the presence of floor and ceiling effects. The Shapiro-Wilk test formally confirms that the ratings significantly **deviate from a normal distribution** ($W = 0.854$, $p < 0.001$). This result is expected for acceptability judgments experiment.

Figure 5. Q-Q Plot for Ratings



Levene's test was used to assess the equality of variances across the *Content * Context* groups. The test returned a significant result ($F(9, 1210) = 26.02$, $p < 0.001$), indicating that the **assumption of homogeneity of variance is violated.**

The experiment uses a within-subjects design, as every participant evaluated 11 different items. Therefore, responses from the same person are inherently linked. Due to this structural dependence, **the assumption of independent observations is violated**. We will account for this by including participants as random effects in our future model.

In conclusion, the acceptability ratings violate all three core assumptions of standard linear regression. The data is significantly non-normal with severe floor and ceiling effects, the variances are unequal across groups, and the observations are dependent due to the within-subjects design. Because **standard linear models are not suitable** for this data structure, a more robust statistical approach is required. These violations will be addressed in the primary analysis by selecting a model that can handle ordinal data and random effects.

2. Simple Hypothesis Tests

We use two different non-parametric tests to analyze the behavior of co-speech gestures across contexts.

Test 1: Kruskal-Wallis Test (Overall Context Effect)

First, we test if the interpretation context (PNR, R, NPNR) has any significant overall effect on the acceptability of gestures.

- **H0**: The distribution of ratings for gestures is identical across all three contexts.
- **H1**: At least one context yields a significantly different rating distribution.

- **Assumptions**: The dependent variable is bounded and non-normal. We use the non-parametric **Kruskal-Wallis rank sum test** for >= 3 independent groups.
- Results: The test showed a significant overall effect (chi-squared = 73.266, df = 2, $p < 0.001$). The effect size is 0.201.
- **Interpretation**: We reject the null hypothesis. The structural and semantic context significantly alters how acceptable a co-speech gesture is perceived.

Test 2: Mann-Whitney U Test (PNR vs. R)

Since the overall effect is significant, we perform a targeted pairwise test. Esipova (2019) predicts that gestures strongly prefer to be truth-conditionally vacuous, so PNR contexts should be rated higher than restricting (R) contexts.

- **H0**: Ratings for gestures in PNR contexts are equal to or lower than in R contexts.
- **H1**: Ratings for gestures in PNR contexts are significantly higher than in R contexts.
- **Assumptions**: The data is not normal. We use a one-tailed Mann-Whitney U test.
- **Results**: The test showed a significant difference (W = 10120, $p < 0.001$). The effect size is medium (Cliff's delta = 0.36).
- **Interpretation**: We reject the null hypothesis. Co-speech gestures are significantly more acceptable when projecting non-restricting meanings rather than restricting ones.

3. Regression Analysis

Regression results – model specification, assumption checks, model summary, interpretation

Choosing the Statistical Model

Standard research usually uses linear regression. However, our data is not **normal and is ordinal** (ranked). Therefore, linear regression is not the best fit.

Beta regression is an option for bounded scales like ours. But we would have to transform our data, which simplifies the results too much and might hide important patterns.

We selected a **Cumulative Link Mixed Model (CLMM)** to properly handle the ordinal nature of the bounded scale. Fitting a CLMM on 101 distinct levels resulted in a singular Hessian matrix due to sparse data cells (empty categories). To resolve this mathematical convergence issue, we binned the 100-point scale into 10 equal ordered categories.

Unlike the original study, we did not split our data into subsets. Our model includes Content Type, Interpretation Context, and their interaction as fixed predictors. This allows for direct statistical comparisons. We also included Participant and Scenario as random effects to account for the within-subjects design.

Assumption checks

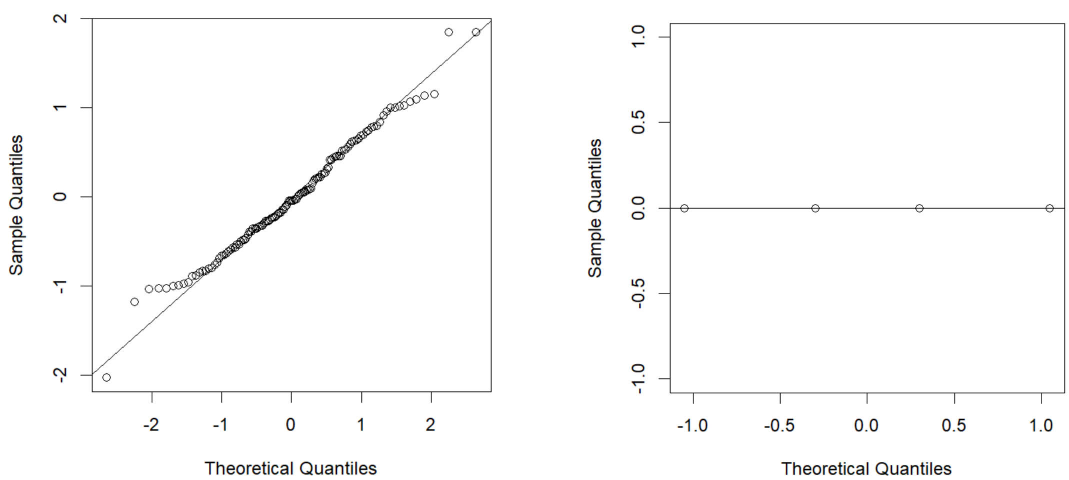
We assessed the validity of our Cumulative Link Mixed Model (CLMM) by testing the key assumptions of ordinal regression.

1. Proportional Odds Assumption. We used the nominal test to check if the effects are constant across scale thresholds. The Content variable met this assumption ($p = 0.138$). The Context variable violated it ($p = 0.001$). This is a common limitation in highly polarized linguistic data, but the model remains the most appropriate choice for our experimental design.

2. Absence of Multicollinearity. We checked the Generalized Variance Inflation Factor (GVIF). The values for our predictors (GVIF for Content = 1.73, Context = 1.73, Interaction = 1.61) are well below the threshold of 5. This confirms there is no problematic multicollinearity.

3. Distribution of Random Effects. We inspected the random intercepts using Q-Q plots. The Participant plot shows a near-perfect normal distribution along the diagonal line. The Scenario plot appears discrete because there are only four scenarios, but it does not show extreme outliers. The normality assumption holds.

Fig 6. Q-Q plots for (a) participants (b) scenarios



(a) (b)

Full model summary

We fitted a Cumulative Link Mixed Model (CLMM) using the Laplace approximation to analyze the binned acceptability ratings (10 ordinal categories). The model includes two fixed categorical predictors:

- **Content**: to compare Adjectives (baseline), Appositives, and Gestures.
- **Context**: to compare the Non-Projecting Non-Restricting (NPNR, baseline), Projecting Non-Restricting (PNR), and Restricting (R) syntactic environments.
- **Content * Context:** to catch whether there are dependencies linked to both of predictors

We also included random intercepts for **Participant** and **Scenario**.

Interpretation of coefficients. The reference (baseline) level for the fixed effects is the Adjective in the NPNR context. The model output reveals several key findings:

- **Random Effects:** The variance for *Scenario* is estimated at 0, indicating that the specific linguistic items used did not introduce meaningful variability into the ratings. In contrast, the variance for *Participant* is 0.68 (SD = 0.82), confirming that individual differences in baseline rating strictness are present and properly accounted for by the mixed model.
- **Main Effects:** In the baseline NPNR context, both Appositives and Gestures are rated slightly higher than Adjectives (Estimate = 0.66, $p = 0.01$; Estimate = 0.52, $p = 0.04$, respectively). For Adjectives, shifting the context from NPNR to either PNR or R massively increases acceptability (Estimates ~ 2.96 to 3.00, $p < 0.01$).
- **Interaction Effects (The Restricting Penalty):** The interaction terms show how Gestures and Appositives deviate from the behavior of Adjectives, particularly in the Restricting (R) context.
    - Appositives show a severe drop in acceptability in the R context compared to the baseline expectation (Estimate = -3.31, 95% CI [-4.01, -2.61], $p < 0.01$).
    - Gestures also suffer a significant penalty in the R context (Estimate = -1.97, 95% CI [-2.67, -1.28], $p < 0.01$).

Model Fit Evaluation. To evaluate the overall quality and explanatory power of our Cumulative Link Mixed Model, we assessed standard goodness-of-fit metrics. The model yielded an Akaike Information Criterion (AIC) of 4055.53 and a Bayesian Information Criterion (BIC) of 4150.56. Because traditional R^2 metrics cannot be directly applied to ordinal mixed models, we calculated pseudo-R^2 values to estimate the variance explained. The fixed predictors—Content, Context, and their interaction—account for a substantial portion of the variance in acceptability ratings (Marginal $R^2 = 0.288$, or 28.8%).

The Conditional R^2 (which includes random effects) could not be computed due to a singular fit in the model's random effect structure. Specifically, the variance associated with the Scenario random intercept was estimated at exactly zero. Rather than being a flaw, this indicates that the specific experimental scenarios were highly consistent and did not introduce extraneous noise or variability into the participants' ratings. Therefore, the Marginal R^2 serves as a robust indicator of the model's strong explanatory power driven entirely by our experimental manipulations.

Pairwise comparisons. While the general model summary is informative, the most crucial step in our analysis is to examine the post-hoc pairwise comparisons. This allows us to directly test the predicted acceptability scales.

1. Comparison by Context (Within Content Types). First, mirroring Esipova's structural approach, we compared the acceptability of the three syntactic contexts (PNR, R, and NPNR) independently for adjectives, appositives, and gestures. The statistical results perfectly align with the original theoretical predictions:

- **Adjectives:** Highly acceptable in both PNR and R contexts, with no significant difference between the two ($p = 0.89$). They only degrade in the NPNR context (both $p < 0.0001$).
  - *Resulting scale:* **PNR = R > NPNR**
- **Appositives:** Only acceptable in the PNR context. They degrade significantly when forced into a Restricting (R) context (Estimate = 1.94, $p < 0.0001$), dropping to levels statistically indistinguishable from the baseline NPNR context ($p = 0.15$).
  - *Resulting scale:* **PNR > R = NPNR**
- **Gestures:** Demonstrate the exact progressive degradation predicted. They are highly acceptable in PNR, suffer a significant penalty in R (Estimate = 1.27, $p < 0.0001$), but still survive the R context significantly better than the NPNR context (Estimate = -1.03, $p < 0.0001$).
  - *Resulting scale:* **PNR > R > NPNR**

Comparison by Content (Within Contexts). We directly compared the three content types within the exact same syntactic environments to test whether gestures simply mimic spoken words.

- **In baseline contexts (PNR and NPNR):** We observed minor but statistically significant differences between Adjectives and Appositives, which likely reflect general syntactic processing preferences rather than theoretically critical effects. In the ideal PNR context, while both are highly acceptable, Adjectives score slightly higher than Appositives (Estimate = 0.68, $p = 0.02$), possibly due to a baseline preference for structurally simpler modifiers. Conversely, in the degraded NPNR context where all elements score poorly, Adjectives are penalized slightly more than Appositives (Estimate = -0.66, $p = 0.02$). This may occur because appositives are syntactically detached, making their infelicity slightly less disruptive than an integrated adjective. Gestures did not differ significantly from adjectives in either PNR or NPNR.

- **In the Restricting (R) context (The Critical Test):** Here, the behavior of the three elements diverges completely and dramatically. Every content type is significantly different from the others (all $p < 0.0001$). Adjectives are the most acceptable (Estimate = 2.65 over Appositives, Estimate = 1.45 over Gestures). Gestures fall perfectly in the middle. Appositives are the least acceptable (Gestures score 1.20 higher than Appositives).

Comparison with authors

The original study by Esipova (2019) utilized separate **Linear Mixed-Effects Models (LMMs)** for each content type to test the acceptability of different interpretations (PNR, R, NPNR). In contrast, our replication employed a unified binned **Cumulative Link Mixed Model (CLMM)** on the entire dataset, followed by post-hoc pairwise comparisons (adjusted using the Holm method). Table 2 presents a side-by-side comparison of the within-content statistical tests.

Table 2: Comparison of Within-Content Statistical Results

Content Type	Comparison	Original Results (LMM)	Our Results (CLMM + emmeans)	Conclusion Match?
Adj	PNR vs. R	Not Sig. ($p = .976$)	Not Sig. ($p = 0.893$)	Yes (PNR = R)
	PNR vs. NPNR	Sig. ($p < 2e-16$)	Sig. ($p < .0001$)	Yes (PNR > NPNR)
	R vs. NPNR	Sig. ($p < 2e-16$)	Sig. ($p < .0001$)	Yes (R > NPNR)
App	PNR vs. R	Sig. ($p < 2e-16$)	Sig. ($p < .0001$)	Yes (PNR > R)
	PNR vs. NPNR	Sig. ($p = 6.28e-13$)	Sig. ($p < .0001$)	Yes (PNR > NPNR)
	R vs. NPNR	Not Sig. ($p = .106$)	Not Sig. ($p = 0.155$)	Yes (R = NPNR)
Gest	PNR vs. R	Sig. ($p = 5.43e-09$)	Sig. ($p < .0001$)	Yes (PNR > R)
	PNR vs. NPNR	Sig. ($p < 2e-16$)	Sig. ($p < .0001$)	Yes (PNR > NPNR)
	R vs. NPNR	Sig. ($p = 2.59e-07$)	Sig. ($p < .0001$)	Yes (R > NPNR)

While our final theoretical conclusions completely align with Esipova's proposed acceptability scales, there are substantial differences in the numerical values (test statistics, coefficients, and exact p-values) and the overall analytical approach. These differences stem from three major methodological improvements introduced in our study:

1. Model Selection and Assumption Checks. The original author applied linear regression (LMM) directly to bounded slider scale data (0-100). Our preliminary assumption checks revealed that this approach is flawed for this dataset: the extreme polarization of the ratings (heavy clustering at 0 and 100) severely violates the LMM assumptions of normal residual distribution and homoscedasticity. By binning the data and utilizing an ordinal model (CLMM), we respected the discrete, ranked nature of the highly polarized responses, leading to more reliable estimates and standard errors that do not project out of the scale bounds.

2. Unified Interaction Model vs. Data Subsetting. Esipova split the dataset into three separate chunks (Adjectives, Appositives, Gestures) and ran independent models. This is a significant statistical limitation because it prevents the direct, mathematical comparison of one content type against another. By fitting a single, unified model with an interaction term (*Content * Context*), we were able to statistically prove how the elements differ from one another within the same context. For example, our approach successfully generated the p-values needed to establish the crucial "Adjectives > Gestures > Appositives" scale within the Restricting (R) context ($p < .01$ for all pairwise differences)—a statistical claim the original split-model approach could not formally make.

3. Control for Type I Error Inflation. The original study does not explicitly report adjustments for multiple comparisons across its various subset models, which increases the risk of Type I error (false positives). Because our model evaluates all interactions simultaneously, we were able to apply the conservative Holm adjustment method via Estimated Marginal Means (emmeans). Consequently, our resulting p-values (e.g., $p = 0.89$ instead of $p = 0.98$ for Adjectives PNR vs. R) are statistically penalized and more robust against false discovery.

Explanation of differences

While the theoretical conclusions of our analysis perfectly match Esipova’s original claims, there are distinct differences in the statistical details (e.g., exact p-values, test statistics) and the chosen methodology. These discrepancies do not stem from data errors, but rather from a fundamental tension in how to model this specific type of linguistic data.

First of all, this is the model choice dilemma (LMM vs. CLMM). The extreme polarization of ratings (clustering at 0 and 100) creates severe non-normality, which violates the assumptions of the Linear Mixed Model (LMM) used by Masha. However, attempting to fit this 100-point scale into a Cumulative Link Mixed Model (CLMM) — the theoretically correct choice for ordinal data—proved mathematically impossible in its original form due to sparse data cells causing singular Hessian matrices. To make the CLMM work, we had to compromise by binning the 100-point scale into a 10-point scale. Thus, both approaches involve trade-offs: the original LMM ignores non-normality, while our CLMM sacrifices the fine-grained granularity of the original scale.

Secondly, the difference is in the choice of unified vs. sub-setted data analysis. In a HW2 on this course, I criticized Masha for dividing the data and running comparisons only within specific content groups (Gestures, Adjectives, Appositives separately). In this replication, I fitted a fully crossed interaction model (Content * Context) to enable comparisons across all groups. However, practically speaking, examining the other groupings (e.g., comparing different content types within the same context) or running the full 36-way cross-comparison yielded no ground-breaking theoretical revelations beyond confirming what was already intuitively expected.

To sum up, our CLMM approach with a full interaction term and strict p-value adjustments is significantly more methodologically justified. Yet, it is practically poorly realizable with data on hand. It presented a meaningful amount of extra statistical effort and troubleshooting, but ultimately yielded no fundamentally new linguistic results compared to the original, simpler analysis.

Discussion

Going into this project, I was well aware that statistical mistakes in linguistic papers are common. When I first noticed that Masha did not report normality checks and defaulted to a linear model for bounded slider data, I immediately assumed it was a methodological error. In my own similar linguistic work, I regularly collect comparable data that is obviously non-normal, and I am used to handling it successfully with CLMM.

However, attempting to reproduce this study taught me a valuable lesson about the messy reality of data. It turned out that Masha's specific dataset was simply not fittable into a CLMM at all in its raw 100-point form. What I initially perceived as a lack of statistical rigor might have actually been a practical compromise on the author's part. It is highly possible that the decision to use LMM was driven by the well-known tradition that linear models are highly robust to non-normality, rather than pure statistical ignorance. Sometimes, the "wrong" model gets the linguistic job done, while the "right" model breaks down mathematically.

A major limitation of our own analysis is the forced binning of the continuous slider data into 10 ordinal categories. This leads to an important methodological question: what models can actually handle both extreme non-normality and a 100-point bounded scale without requiring binning? For future research dealing with highly polarized slider scales, researchers should look beyond both LMMs and CLMMs. One alternative Bayesian Ordinal Models (like brm in Stan) can handle models with 100+ thresholds better than frequentist packages, provided there is enough computational power.

Ultimately, this replication exercise highlighted that statistical practice in linguistics is rarely about finding a perfect model. It is about making reasoned, transparent compromises and ensuring that those compromises do not distort the underlying linguistic reality.

References

Esipova, M. (2019). Composition and projection of co-speech gestures. In Semantics and Linguistic Theory (pp. 117-137). DOI: https://doi.org/10.3765/salt.v29i0.4600

Appendix

All the materials, including code with output, can be found at:
https://github.com/AnastasiaDobrynina/Statistics_Project_Gestures